7.4

$f_{X+Y}(z) = f_X * f_Y(z) = \int_0^{\infty} f_X(x) f_Y(z-x) dx$

$\int_0^z \lambda e^{-\lambda x} * \mu e^{-\mu(z-x)} dx$

$\int_0^z \lambda \mu e^{-\lambda x + \mu x - \mu z} dx$

$\int_0^z \lambda \mu e^{(-\lambda+\mu)x - \mu z} dx$

$\lambda\mu(\frac{1}{\mu-\lambda} e^{(\mu-\lambda)x - \mu z})|_0^z$

$\frac{\lambda\mu}{\mu-\lambda} e^{-\mu z}(e^{(\mu-\lambda)x})|_0^z$

$\frac{\lambda\mu}{\mu-\lambda} e^{-\mu z}(e^{(\mu-\lambda)z} - 1)$

$\frac{\lambda\mu}{\mu-\lambda}(e^{-\lambda z} - e^{-\mu z})$ for $z \geq 0$

$f_{X+Y}(z) = \frac{\lambda\mu}{\mu-\lambda}(e^{-\lambda z} - e^{-\mu z})$ for $z \geq 0$ else 0

7.6

The positions are symmetric so we can use the exchangability rule.

$P(X_3 = K, X_5 = A_S) = P(X_1 = K, X_2 = A_S) = \frac{4}{52}\frac{1}{51} = \frac{1}{663}$

Alternatively: $\forall i, j(i \neq j) \to P(X_i = A_S, X_j = K_S) = \frac{50!}{52!} = \frac{1}{52 \times 51}$

Since there are 4 different possibilities we need to look out for $4 \times P(X_i = A_S, X_j = K_S) = \frac{1}{663}$

7.16

$p_{X+Y}(n) = p_X * p_Y(n) = \sum_k p_X(n-k) p_Y(n)$

$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}$

$p_Y(k) = p^k(1-p)^{1-k}$ for $k \in \{0, 1\}$

$\sum_{k=0}^{1} p^k(1-p)^{1-k} \frac{\lambda^{n-k} e^{\lambda}}{(n-k)!}$

$(1-p)\frac{\lambda^n e^{\lambda}}{n!} + p\frac{\lambda^{n-1} e^{-\lambda}}{(n-1)!}$

$e^{-\lambda}((1-p)\frac{\lambda^n}{n!} + p\frac{\lambda^{n-1}}{(n-1)!})$

For $n = 0 \rightarrow (1-p)e^{-\lambda}$

For $n > 0 \rightarrow \frac{e^{-\lambda}\lambda^{n-1}}{(n-1)!}((1-p)\frac{\lambda}{n} + p)$

$$p_{X+Y}(n) = \begin{cases} 0 & n < 0 \\ (1-p)e^{-\lambda} & n = 0 \\ \frac{e^{-\lambda}\lambda^{n-1}}{(n-1)!}((1-p)\frac{\lambda}{n} + p) & n > 0 \end{cases}$$

7.22

$X, Y \sim N(\mu, \sigma^2)$

$2X \sim N(2\mu, (2\sigma)^2) \rightarrow 2X \sim N(2\mu, 4\sigma^2)$

$p_{X+Y}(n) = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(n-x-\mu)^2}{2\sigma^2}} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$

$p_{X+Y}(n) = \frac{1}{\sigma^2 2\pi} \int_{-\infty}^{\infty} e^{-\frac{(n-x-\mu)^2-(x-\mu)^2}{2\sigma^2}} dx$

Evaluated with integral calculator

$p_{X+Y}(n) = \frac{1}{2\sigma\sqrt{\pi}} e^{-\frac{(n-2\mu)^2}{4\sigma^2}}$

According to this pdf, $X + Y \sim N(2\mu, 2\sigma^2)$

Thus the parameter σ^2 for $2X$ and $X + Y$ are not the same for $\sigma^2 > 0$ while they still have the same mean of 2μ

7.28

$\int_0^{\infty} \int_{x_1}^{\infty} \int_{x_2}^{\infty} \lambda e^{-\lambda x_1} \lambda e^{-\lambda x_2} \lambda e^{-\lambda x_3} dx_3 dx_2 dx_1$

$\int_0^{\infty} \int_{x_1}^{\infty} \int_{x_2}^{\infty} \lambda^3 e^{-\lambda(x_1+x_2+x_3)} dx_3 dx_2 dx_1$

$\int_0^{\infty} \int_{x_1}^{\infty} \lambda^2 e^{-\lambda(x_1+2x_2)} dx_2 dx_1$

$\int_0^{\infty} \frac{\lambda}{2} e^{-\lambda(3x_1)} dx_1$

$-\frac{1}{6} e^{-\lambda(3x_1)}|_0^{\infty} = \frac{1}{6}$

8.2

$E(X_4 + X_6 + X_{12}) = E(X_4) + E(X_6) + E(X_{12}) = 2.5 + 3.5 + 6.5 = 12.5$

8.8

$E(X_p + X_f) = E(X_p) + E(X_f) = 4 + 0.5 = 4.5 \text{ PM}$

$Var(X_p + X_f) = Var(X_p) + Var(X_f) = \frac{1}{12}(6)^2 + 0.5^2 = 3 + \frac{1}{4} = 3.25$